```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import OrdinalEncoder,KBinsDiscretizer
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import CategoricalNB
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

df=pd.read_csv("adult.csv")
df
```

```
       age workclass  fnlwgt     education  education.num
marital.status  \
0       90         ?   77053       HS-grad              9
Widowed
1       82   Private  132870       HS-grad              9
Widowed
2       66         ?  186061  Some-college             10
Widowed
3       54   Private  140359       7th-8th              4
Divorced
4       41   Private  264663  Some-college             10
Separated
...    ...       ...     ...           ...            ...
...
32556   22   Private  310152  Some-college             10       Never-
married
32557   27   Private  257302    Assoc-acdm             12  Married-
civ-spouse
32558   40   Private  154374       HS-grad              9  Married-
civ-spouse
32559   58   Private  151910       HS-grad              9
Widowed
32560   22   Private  201490       HS-grad              9       Never-
married

              occupation   relationship   race     sex
capital.gain  \
0                      ?  Not-in-family  White  Female             0

1        Exec-managerial  Not-in-family  White  Female             0

2                      ?      Unmarried  Black  Female             0

3      Machine-op-inspct      Unmarried  White  Female             0

4         Prof-specialty      Own-child  White  Female             0
```

```
...                  ...            ...    ...     ...           ...

32556    Protective-serv  Not-in-family  White    Male             0

32557       Tech-support           Wife  White  Female             0

32558  Machine-op-inspct        Husband  White    Male             0

32559       Adm-clerical      Unmarried  White  Female             0

32560       Adm-clerical      Own-child  White    Male             0


       capital.loss  hours.per.week native.country income
0              4356              40  United-States  <=50K
1              4356              18  United-States  <=50K
2              4356              40  United-States  <=50K
3              3900              40  United-States  <=50K
4              3900              40  United-States  <=50K
...             ...             ...            ...    ...
32556             0              40  United-States  <=50K
32557             0              38  United-States  <=50K
32558             0              40  United-States   >50K
32559             0              40  United-States  <=50K
32560             0              20  United-States  <=50K

[32561 rows x 15 columns]
```

```
features=['age','education.num','marital.status','capital.gain','capit
al.loss','hours.per.week']
X=df[features]
y=df['income']

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,rando
m_state=1172)

num_cols=['age','capital.gain','capital.loss','hours.per.week']
cat_cols=['marital.status']
passthrough_cols=['education.num']

preprocessor =ColumnTransformer(
    transformers=[

('num',KBinsDiscretizer(n_bins=5,encode='ordinal',strategy='quantile')
,num_cols),
        ('cat',OrdinalEncoder(),cat_cols),
        ('pass','passthrough',passthrough_cols)
    ]
)
```

```
X_train_processed=preprocessor.fit_transform(X_train)
```

```
/Users/sparshkarna/dev/sem6/ml-lab/myenv/lib/python3.9/site-packages/
sklearn/preprocessing/_discretization.py:306: UserWarning: Bins whose
width are too small (i.e., <= 1e-8) in feature 1 are removed. Consider
decreasing the number of bins.
  warnings.warn(
/Users/sparshkarna/dev/sem6/ml-lab/myenv/lib/python3.9/site-packages/
sklearn/preprocessing/_discretization.py:306: UserWarning: Bins whose
width are too small (i.e., <= 1e-8) in feature 2 are removed. Consider
decreasing the number of bins.
  warnings.warn(
/Users/sparshkarna/dev/sem6/ml-lab/myenv/lib/python3.9/site-packages/
sklearn/preprocessing/_discretization.py:306: UserWarning: Bins whose
width are too small (i.e., <= 1e-8) in feature 3 are removed. Consider
decreasing the number of bins.
  warnings.warn(
```

```
X_test_processed = preprocessor.transform(X_test)
```

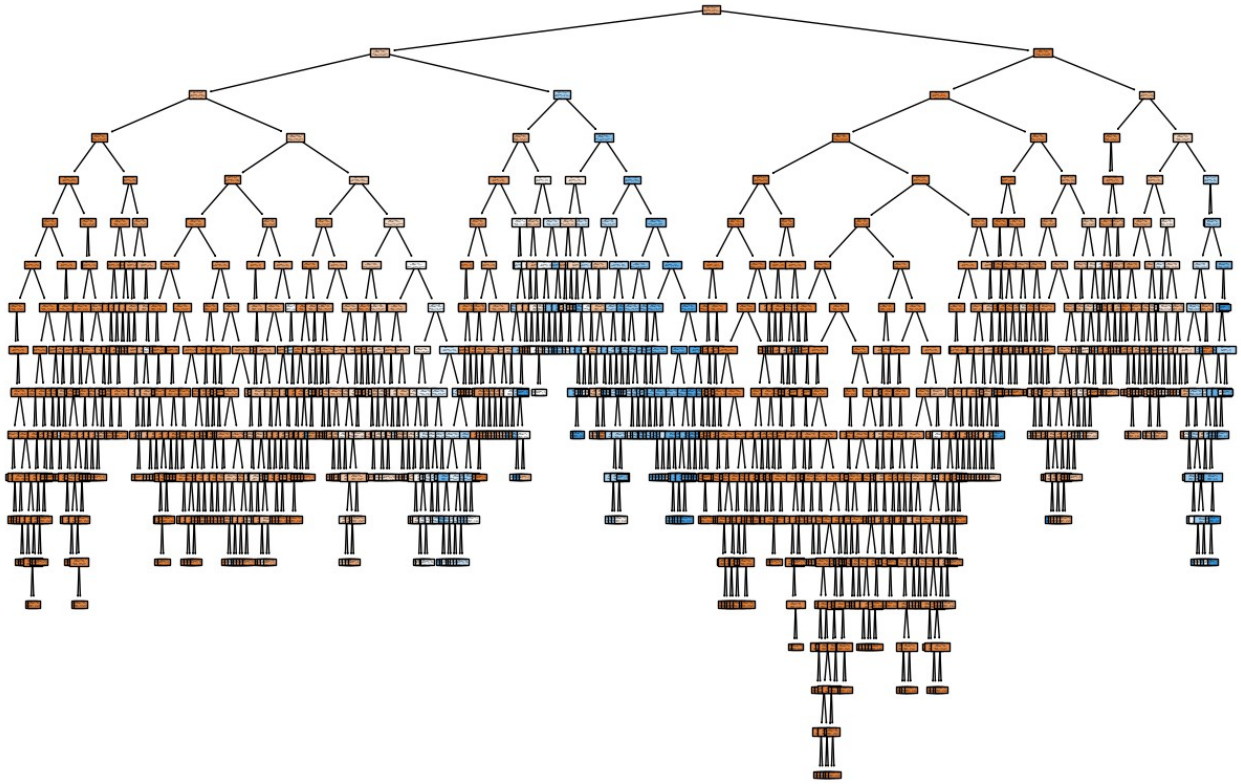
```
cart_model = DecisionTreeClassifier(criterion='gini',
random_state=1172)
cart_model.fit(X_train_processed, y_train)
```

```
DecisionTreeClassifier(random_state=1172)
```

```
plt.figure(figsize=(15, 10))
plot_tree(cart_model,

feature_names=['age','education.num','marital.status','capital.gain','
capital.loss','hours.per.week'],
          class_names=['<=50K', '>50K'],
          filled=True)

plt.show()
```



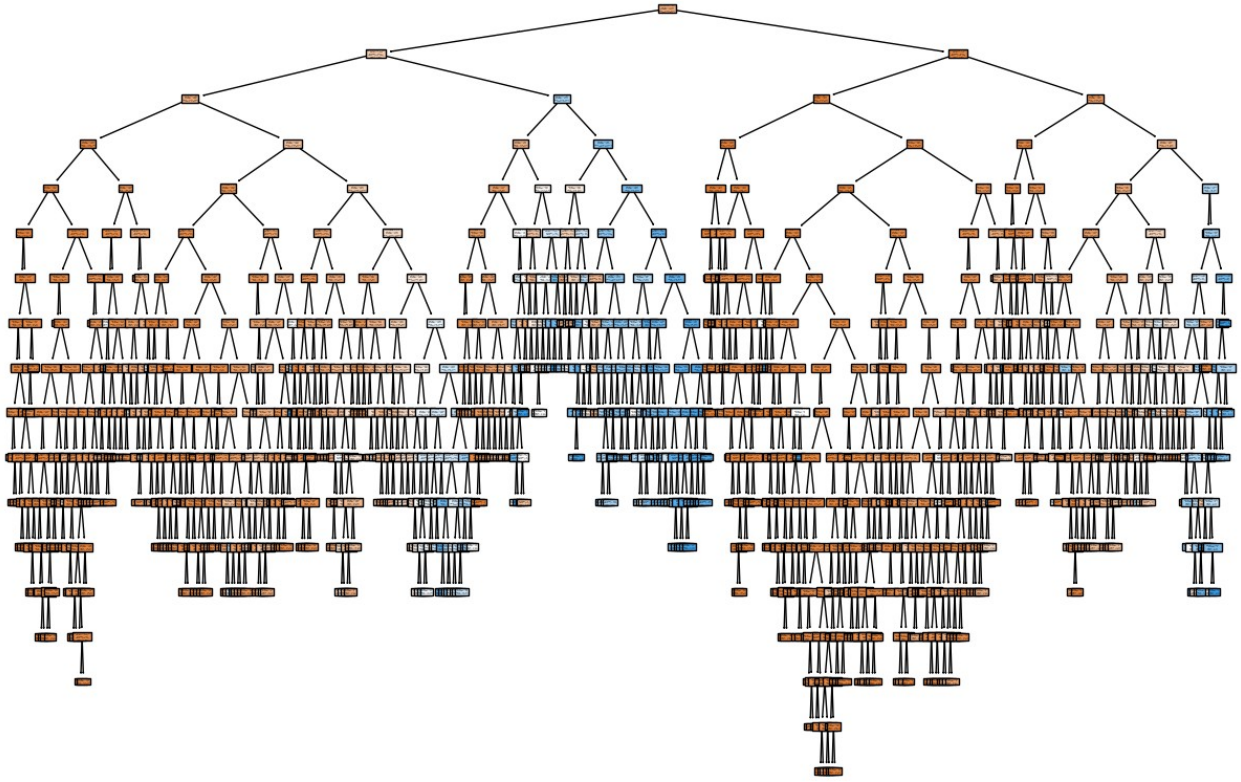
```
id3_model = DecisionTreeClassifier(criterion='entropy',
random_state=1172)
id3_model.fit(X_train_processed, y_train)
```

```
DecisionTreeClassifier(criterion='entropy', random_state=1172)
```

```
plt.figure(figsize=(15, 10))
plot_tree(id3_model,

feature_names=['age','education.num','marital.status','capital.gain','
capital.loss','hours.per.week'],
          class_names=['<=50K', '>50K'],
          filled=True)

plt.show()
```



```
nb_model = CategoricalNB()
nb_model.fit(X_train_processed, y_train)
```

```
CategoricalNB()
```

```
y_pred_cart = cart_model.predict(X_test_processed)
accuracy_score(y_test, y_pred_cart)
```

```
0.8216808271061521
```

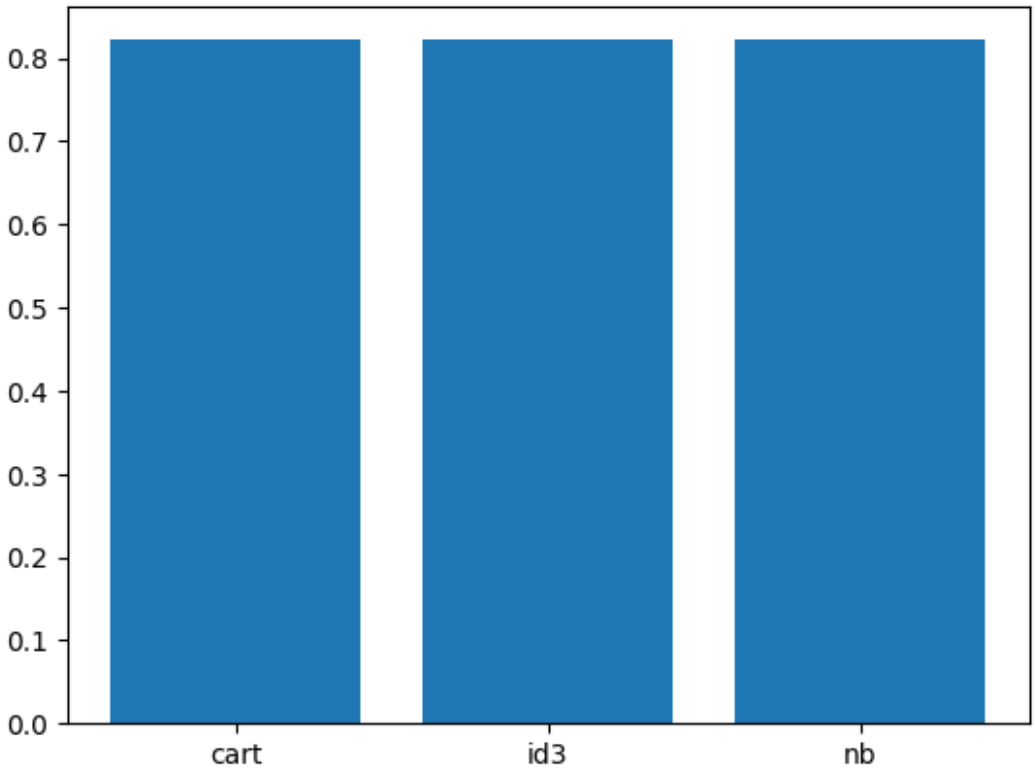
```
y_pred_id3 = id3_model.predict(X_test_processed)
accuracy_score(y_test, y_pred_id3)
```

```
0.8217831917289384
```

```
y_pred_naive = nb_model.predict(X_test_processed)
accuracy_score(y_test, y_pred_naive)
```

```
0.8215784624833657
```

```
plt.bar(["cart", "id3", "nb"], [accuracy_score(y_test, y_pred_cart),
accuracy_score(y_test, y_pred_id3), accuracy_score(y_test,
y_pred_naive)])
plt.show()
```


0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
cart
id3
nb